Laboratory 3 Report

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This report presents a comparative study of anomaly detection methods for network intrusion detection. We evaluate and compare four main approaches: One-Class SVM, autoencoder-based reconstruction, hybrid dimensionality reduction techniques, and unsupervised clustering algorithms. The models are assessed under a train-test distribution shift, with the aim of identifying anomalous patterns in unseen network traffic. Additionally, we investigate the impact of training data contamination on model robustness by introducing varying levels of anomalous traffic during the training phase.

1 INTRODUCTION

This laboratory investigates a spectrum of anomaly detection techniques — from shallow one-class classifiers to deep autoencoders and purely unsupervised clustering — applied to a network traffic dataset derived from the KDD Cup 1999 benchmark. The dataset contains 43 features describing individual network connections, with labels indicating the traffic type (Normal, DoS, Probe, R2L). Our analysis is structured around four main tasks:

(1) **Dataset characterization and preprocessing** (Section 2): We explore the dataset structure, handle categorical and numerical features, and analyze per-class feature distributions through heatmaps.
(2) **Shallow anomaly detection** (Section 3): We evaluate One-Class SVM under three training configurations: normal data only (novelty detection), all data (outlier detection), and mixed contamination levels. We quantify the impact of the ν parameter and training data composition.
(3) **Deep anomaly detection and data representation** (Section 4): We train an autoencoder exclusively on normal data and explore three detection strategies: reconstruction error thresholding, OC-SVM on bottleneck embeddings, and OC-SVM on PCA-reduced features.
(4) **Unsupervised anomaly detection and interpretation** (Section 5): We apply K-Means and DBSCAN to assess whether purely unsupervised approaches can identify attack patterns without any label supervision, using silhouette analysis, t-SNE visualization, and cluster purity metrics.

2 DATASET CHARACTERIZATION AND PREPROCESSING

This section describes the dataset, the preprocessing pipeline, and the exploratory analysis performed to understand the feature space and identify discriminative patterns.

2.1 Dataset Exploration

The dataset consists of network traffic logs derived from the KDD Cup 1999 benchmark, provided as separate training and test CSV files. Each record represents a single network connection, described by 43 columns: 41 features, a multi-class attack label, and a binary label (`normal` vs. `anomaly`). Of the 41 features, **3 are categorical** (`protocol_type`, `service`, `flag`), describing the connection protocol, destination service, and TCP status flag respectively. The remaining **38 are numerical** and capture intrinsic connection properties, such as duration and transferred bytes, alongside time-windowed traffic metrics like connection rates and error percentages.

Class	Train (80%)	Val (20%)	Test	Binary Label
Normal	10,758 (71.41%)	2,690 (71.41%)	2,152 (36.94%)	Normal
DoS	2,330 (15.47%)	583 (15.47%)	2,577 (44.23%)	Anomaly
Probe	1,830 (12.15%)	458 (12.15%)	1,097 (18.83%)	Anomaly
R2L	145 (0.96%)	36 (0.96%)	0 (0.0%)	Anomaly
Total	15,064	3,767	5,826	

Table 1. Label distribution across dataset splits.

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As detailed in Table 1, the training set contains 18,831 samples, with approximately 71.41% normal traffic and 28.59% anomalies. The test set contains 5,826 samples with a markedly different composition: only 36.94% normal versus 63.06% anomalous traffic. This shift creates an intentional domain mismatch that stress-tests model generalization beyond training conditions. Specifically, the R2L attack class is entirely missing from the test set, while the prevalence of DoS attacks nearly triples from 15.47% in the training data to 44.23% in the test data.

To generate the validation set shown in Table 1, we perform an 80/20 stratified split. This yields a training partition of 15,064 samples and a validation partition of 3,767 samples. The rationale for isolating validation data prior to preprocessing is detailed in Section 2.2.

2.2 Preprocessing Pipeline

The preprocessing pipeline is designed to ensure clean, standardized input for all downstream models while strictly preventing data leakage.

Data cleaning. We checked for missing values (`NaN`, `Inf`) and duplicates, finding one duplicate row which was removed. We then dropped four zero-variance columns (`land`, `urgent`, `num_outbound_cmds`, `is_host_login`), reducing the feature space from 41 to 37 attributes.

Train/validation split. The 80/20 stratified split was performed *before* any feature transformation. This strategy guarantees that the scaler and encoder are fitted exclusively on the training data. By strictly separating the partitions early in the pipeline, we simulate a realistic scenario where validation data is entirely unseen, thereby preventing any statistical information from leaking into the validation process and ensuring unbiased performance estimates.

Categorical encoding. Rare values below a frequency threshold were grouped into an `other` category. We retained the 8 most frequent services, 5 flags, and all 3 protocols, then applied One-Hot Encoding with `handle_unknown='ignore'` to handle unseen categories in the test set gracefully.

Numerical standardization. All numerical features were standardized using `StandardScaler`, fitted exclusively on the training partition and applied identically to validation and test sets. After preprocessing, the final feature space consists of **51 dimensions**: 34 numerical features and 17 one-hot encoded categorical dimensions.

2.3 Domain Analysis: Feature Heatmaps

To understand which features characterize each attack type, we computed per-class heatmaps of the standardized mean, standard deviation, and median feature values (Figure 1).

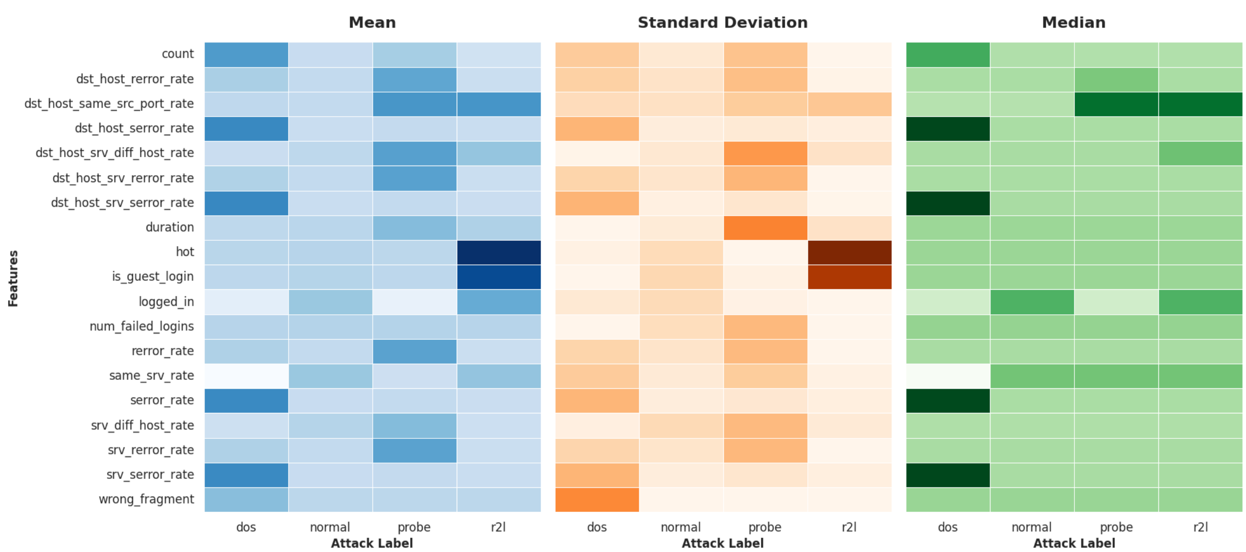

Fig. 1. Feature analysis across attack classes: standardized Mean, Standard Deviation, and Median. DoS attacks exhibit strong error-rate signals, while R2L attacks closely mimic normal traffic.

The heatmaps reveal clear discriminative structure, highlighting key features that differentiate the traffic classes. **DoS attacks** exhibit extremely elevated `serror_rate` and `srv_serror_rate` ($\mu > 2.0$ standard deviations above the global mean), reflecting SYN flood behavior, alongside high `count` and `srv_count` values consistent with connection floods. Additionally, DoS attacks show a significant negative mean/median for `same_srv_rate`, indicating that attackers hit many different services rather than concentrating on a single service as is common in benign traffic, and elevated values for `wrong_fragment` due to malformed packet transmission. **Probe attacks** are characterized by high `dst_host_srv_diff_host_rate` and elevated service diversity metrics, reflecting systematic port and host scanning. They also display elevated values for `dst_host_same_src_port_rate`, indicating systematic scanning from a single port. Notably, the medians for these features are considerably lower than their corresponding means, illustrating a highly skewed distribution driven by a few intense, extreme scanners. **R2L attacks** show only subtle signatures — slightly elevated `hot` and `is_guest_login` — but otherwise closely resemble normal traffic, explaining why they are the hardest class to detect. **Normal traffic** exhibits low, stable values across all anomaly-indicative features with moderate connection rates, and is uniquely characterized by high values in `logged_in` and `service_http`, which are the most distinctive signatures of benign, legitimate user activity.

2.4 Feature Correlation Analysis

To quantify feature redundancy, we computed the Pearson correlation matrix (unfortunately not shown here due to space constraints) on the numerical training features. Several feature groups exhibit near-perfect correlation (e.g., `serror_rate` and `srv_serror_rate`: $r > 0.98$; `dst_host_serror_rate` and `dst_host_srv_serror_rate`), confirming that the feature space contains significant redundancy. This motivates PCA-based dimensionality reduction in Section 4.

3 SHALLOW ANOMALY DETECTION

In this section, we investigate One-Class Support Vector Machine (OC-SVM) for anomaly detection under three training configurations: normal data only (novelty detection), all data (outlier detection), and mixed data with varying anomaly contamination ratios.

3.1 OC-SVM with Normal Data Only (Novelty Detection)

In novelty detection, the OC-SVM is trained exclusively on normal traffic, learning a tight boundary around the normal data manifold. The key hyperparameter is ν, which controls the upper bound on the fraction of training errors. Since the training data is purely normal, ν should reflect the expected fraction of noisy or edge-case normal samples. We evaluated four values: $\nu \in \{0.001, 0.01, 0.05, 0.5 \text{ (default value)}\}$.

ν	**Train F1-Macro**	**Test F1-Macro**
0.001	0.9058	0.7382
0.01	0.9039	0.7401
0.05	**0.9026**	**0.7477**
0.5 (default)	0.6378	0.4647

Table 2. OC-SVM (Normal-Only): impact of ν on performance.

We selected $\nu = 0.05$ as the optimal value, as it achieves the highest test F1-Macro (0.7477), allowing the model to exclude approximately 5% of normal samples as edge-case noise and creating a slightly looser boundary that generalizes better. However, the variation in performance among the three lower values tested ($\nu \in \{0.001, 0.01, 0.05\}$) is minimal. This highlights that the ν parameter is not a strict limit, but rather an upper boundary to the number of anomalies the model expects. The default value $\nu = 0.5$ forces instead 50% of normal training samples to lie outside the decision boundary, systematically undermining the normality model and collapsing test F1-Macro to 0.4647.

3.2 OC-SVM with All Data (Outlier Detection)

When training on the full dataset (normal + anomalies), the OC-SVM operates as an *outlier detector*. We set ν to the true contamination rate: $\nu = 4305/15063 \approx 0.286$. The Normal-Only model outperforms the All-Data model by approximately **10 percentage points** in test F1-Macro (0.7477 vs. 0.6484), demonstrating that novelty detection yields a more effective decision boundary than outlier detection on contaminated data.

3.3 Sensitivity Analysis: Impact of Anomaly Contamination

To quantify the effect of training data contamination, we trained OC-SVMs with increasing proportions of anomalous data (0%, 10%, 20%, 50%, 100%), setting ν proportionally to the anomaly ratio at each level.

Anomaly %	ν	Train F1-Macro	Test F1-Macro
0% (Normal-Only)	0.01	**0.9039**	**0.7401**
10%	0.038	0.6628	0.5137
20%	0.074	0.6779	0.5589
50%	0.167	0.7226	0.5975
100% (All-Data)	0.286	0.7364	0.6482

Table 3. Sensitivity analysis: impact of anomaly contamination on OC-SVM performance.

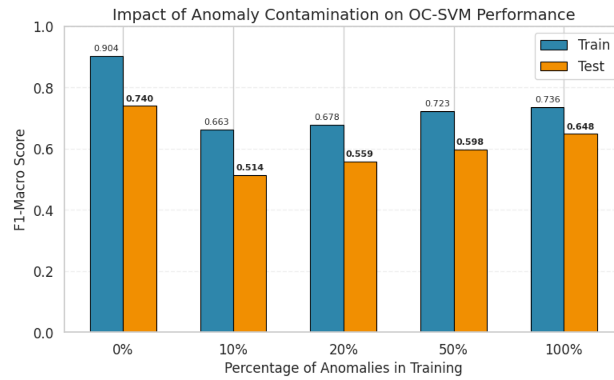

Fig. 2. F1-Macro scores across anomaly contamination levels.

The results exhibit a **non-monotonic relationship** between contamination level and test F1-Macro: performance degrades sharply at 10% contamination (F1 = 0.514) before partially recovering as contamination reaches 100% (F1 = 0.648). This recovery reflects ν growing large enough to absorb the anomaly fraction, though no contaminated configuration approaches the clean baseline (F1 = 0.740). Small contamination levels are the most damaging because the model receives conflicting signals about what constitutes normal behavior, yet ν remains too small to properly account for the anomalies present.

3.4 Robustness Evaluation

We compared three representative models on the test set to assess generalization and per-attack detection robustness.

Model	Train F1	Test F1	Most Missed (FNR%)
Normal-Only (ν=0.05)	0.9026	**0.7477**	DoS (23.6%)
All-Data (ν=0.286)	0.7364	0.6484	DoS (41.7%)
10% Anomalies	0.6628	0.5137	DoS (77.2%)

Table 4. Robustness evaluation on test set, including Train F1 and per-attack False Negative Rate (FNR).

The Normal-Only model (ν = 0.05) achieves the best performance on both the training and test set (Table 4). This confirms that the model generalizes pretty well and does not overfit to specific training anomalies, while the other two models, trained on contaminated data, show a significant performance drop.

On the test set, errors are split between False Positives (normal traffic flagged as anomalous due to tight boundaries) and False Negatives (missed anomalies). Per-attack analysis on the Normal-Only model (ν = 0.05) reveals that DoS attacks are the hardest to detect, with 609 of 2,577 samples misclassified (FNR = 23.6%). This evasion occurs because DoS patterns (especially low-rate attacks) closely mimic legitimate high-traffic scenarios. Conversely, Probe attacks are highly identifiable, achieving an FNR of 2.8% (31 of 1,097 samples missed) since scanning and reconnaissance activities diverge clearly from normal network patterns.

Training contamination is particularly damaging for DoS detection: the FNR rises to 41.7% in the All-Data model and peaks at 77.2% with 10% contamination. This represents a **3.3× increase in missed DoS attacks** from minimal training pollution, highlighting how conflicting signals degrade decision boundaries. While the subtle signatures of R2L attacks make them theoretically the most challenging class to detect, their complete absence from the test set prevents any empirical evaluation of their robustness. Nevertheless, on the evaluated classes, the Normal-Only approach achieves the best precision-recall balance, confirming that novelty detection is the most robust strategy when clean data is available.

4 DEEP ANOMALY DETECTION AND DATA REPRESENTATION

This section explores deep learning and dimensionality reduction approaches for anomaly detection: autoencoder-based reconstruction error thresholding, autoencoder bottleneck embeddings with OC-SVM, and PCA-based dimensionality reduction with OC-SVM.

4.1 Autoencoder Architecture and Training

We designed a symmetric autoencoder with the following architecture:

- **Encoder:** Input (51) $\longrightarrow$ 128 $\longrightarrow$ 64 $\longrightarrow$ 16 (bottleneck), with BatchNorm, ReLU, and Dropout (0.2) after each h layer.
- **Decoder:** 16 $\longrightarrow$ 64 $\longrightarrow$ 128 $\longrightarrow$ 51 (reconstruction), with a symmetric architecture.

Since the feature space contains both numerical and categorical (one-hot encoded) features, we used a **composite loss function**: MSE for numerical features and Cross-Entropy for each categorical feature. This ensures the autoencoder learns appropriate reconstruction objectives for each feature type. The model was trained exclusively on the **normal data** from the training partition (10,758 samples) in a semi-supervised way: by learning to reconstruct only normal traffic, anomalies should produce higher reconstruction errors due to unfamiliar feature patterns.

4.2 Hyperparameter Tuning

We performed a grid search over learning rates $\{5 \times 10^{-3},\ 10^{-3},\ 5 \times 10^{-4},\ 10^{-4}\}$ with a maximum of 150 epochs and early stopping (patience = 15).

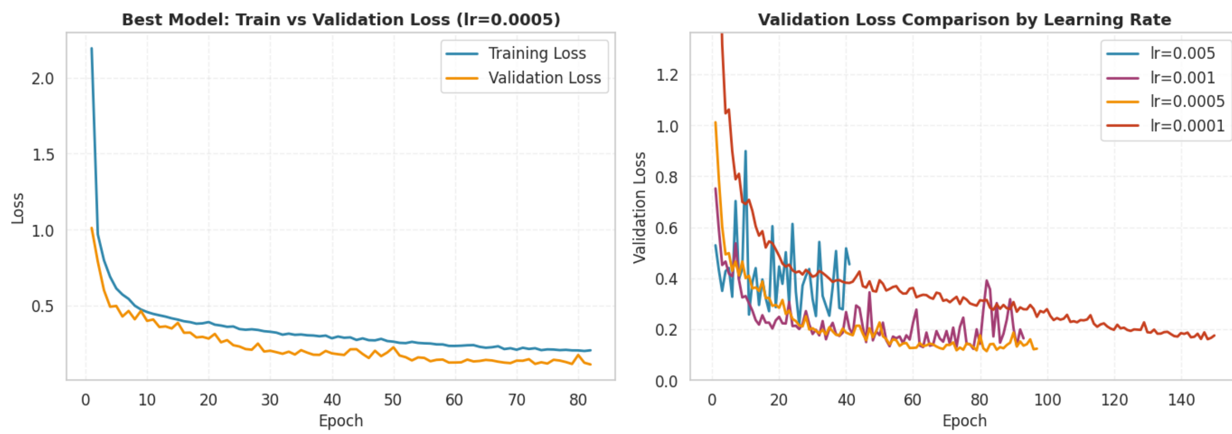

Fig. 3. AE validation loss curves across four learning rates (right) and train and val curves for the best model (left).

The learning rate 5×10^{-4} achieves the lowest validation loss (0.1140) with a stable convergence. Higher learning rates converge faster but to worse optima due to instability in the early training phase, while 10^{-4} reaches an higher minimum and slower convergence.

4.3 Reconstruction Error Analysis

We computed the Empirical Cumulative Distribution Function (ECDF) of reconstruction errors on the autoencoder validation partition containing normal traffic only. An analysis of the underlying error statistics justified this choice: while the distribution possesses a tight core (Median = 0.0117 and Mean = 0.1140), it exhibits a heavy right tail stretching to an absolute maximum error of 44.7958. This confirms that a small subset of normal traffic generates high reconstruction errors, likely due to variations in traffic patterns or slight model misspecification. Consequently, the 95th percentile was selected as the operational anomaly threshold: $\theta = 0.3604$. This design choice explicitly establishes a baseline False Positive Rate (FPR) of approximately 5% on pure normal traffic, balancing detection sensitivity against the operational costs of false alarms.

This threshold choice is visually supported by Figure 4 (a), where the validation ECDF flattens past 1.0 as it enters the outlier tail. In Figure 4 (b), clear rightward shifts validate the model's logic: while the normal-only validation set has 95% of samples below θ, the training set crosses the threshold at 71%, reflecting the 29% anomaly contamination. The test set shifts furthest to the right, crossing the threshold at approximately 32%, which aligns with its 63% anomaly density and confirms that unseen attack signatures produce high reconstruction errors.

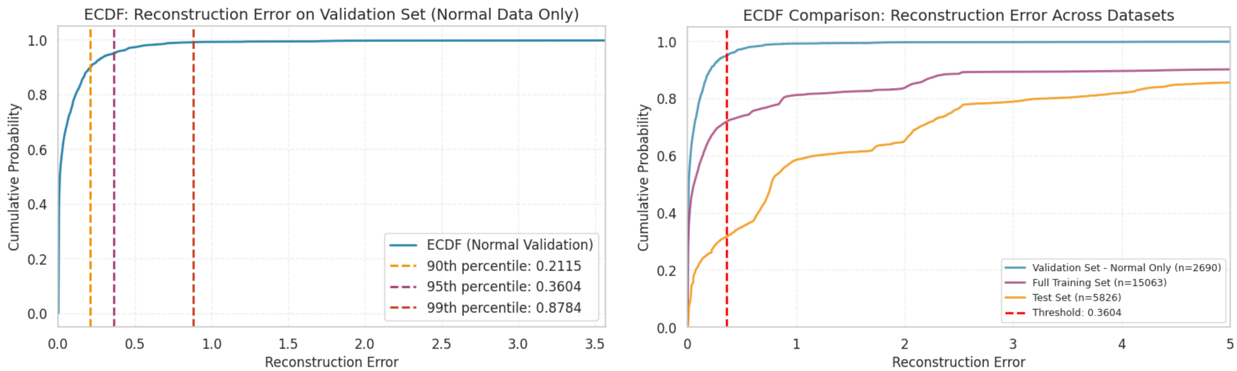

Fig. 4. Empirical Cumulative Distribution Function (ECDF) analysis of Autoencoder reconstruction errors. Figure (a) shows the localized threshold selection on the normal-only Validation split, highlighting candidate percentiles. Figure (b) demonstrates the structural rightward shift of the anomaly-laden Training and Test curves relative to the clean validation manifold.

Samples with a reconstruction error exceeding θ are classified as anomalies. As shown in Table 5, the model achieves a test F1-Macro of **0.7547**, demonstrating strong generalization and confirming the efficacy of the learned normal traffic manifold. Note that performance metrics on the Validation Set in Table 5 utilize the full 3,767-sample stratified validation partition (which includes anomalies) to evaluate threshold efficacy. In contrast, the autoencoder's weights were trained, and the threshold θ itself was selected, using exclusively the 2,690 normal-traffic samples within that partition.

Dataset	F1-Macro	Precision (Anomaly)	Recall (Anomaly)
Train Set	0.9132	0.8823	0.8690
Validation Set	0.9024	0.8713	0.8487
Test Set	**0.7547**	0.7994	0.8669

Table 5. Autoencoder reconstruction error anomaly detection ($\theta = 0.3604$).

4.4 Latent Space Anomaly Detection (AE + OC-SVM)

We extracted the 16-dimensional bottleneck embeddings from the trained encoder and used them as input features for an OC-SVM trained on normal data only. We tested two values of ν: 0.01 (conservative) and 0.1. The choice of $\nu = 0.1$ was empirically motivated: the ECDF of reconstruction errors for normal samples in the bottleneck space shows a roughly 10th-percentile inflection, suggesting that approximately 10% of normal points inhabit atypical regions of the compressed manifold. Setting ν below this fraction forces the OC-SVM to exclude geometrically valid normal points, shrinking the boundary and inflating false positives.

ν	Train F1-Macro	Test F1-Macro
0.01	0.7662	0.6645
0.1	**0.8300**	**0.7795**

Table 6. AE Bottleneck + OC-SVM: impact of ν on performance.

Dataset Partition	Precision	Recall	F1-Macro
Train Set	0.7529	0.7628	0.8300
Validation Set	0.7588	0.7391	0.8250
Test Set	**0.8168**	**0.8821**	**0.7795**

Table 7. Performance metrics for the optimal AE Bottleneck + OC-SVM configuration ($\nu = 0.1$).

The $\nu = 0.1$ configuration yields a substantial **+11.5 percentage points** improvement on the test set compared to the $\nu = 0.01$ baseline, making AE Bottleneck + OC-SVM the **best-performing method overall** (test F1-Macro = 0.7795). This improvement occurs because the conservative $\nu = 0.01$ is too restrictive in the compressed 16-dimensional latent space, where normal traffic embeddings exhibit greater relative dispersion and require a looser boundary to capture legitimate variations. By allowing this wider boundary, the optimal model achieves an exceptional anomaly recall of **88.21%** while maintaining a strong precision of **81.68%**, demonstrating that non-linear feature compression effectively isolates structural anomalies while mitigating the risk of alert fatigue.

4.5 PCA-based Anomaly Detection

As a linear baseline for dimensionality reduction, we applied Principal Component Analysis (PCA) strictly on the normal training data. Analyzing the cumulative explained variance allowed us to determine the optimal sub-space projection; as illustrated in Figure 5, an explicit elbow point emerges at **20 components**, capturing exactly 95.5% of the total dataset variance. This confirms the significant feature redundancy mathematically identified in our earlier correlation analysis (Section 2.4). By reducing the input space from 51 to 20 dimensions, we achieve a substantial **60% dimensionality reduction** while retaining the vast majority of information. An OC-SVM ($\nu = 0.01$) was subsequently trained on this optimized linear sub-space.

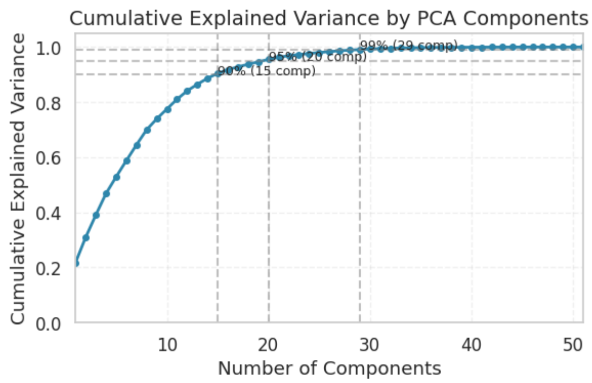

Fig. 5. Cumulative explained variance vs. number of PCA components (fitted on normal training data only). The elbow at 20 components (95.5% cumulative variance) guides our selection.

Dataset	F1-Macro	Precision (A)	Recall (A)
Train Set	0.8835	0.9655	0.7206
Validation Set	0.8664	0.9548	0.6862
Test Set	**0.7377**	**0.8533**	**0.7186**

Table 8. PCA + OC-SVM performance (20 components).

The PCA + OC-SVM pipeline achieves a final test F1-Macro of 0.7377 (Table 8), nearly matching our full-dimensional raw baseline (0.7477) despite working with less than half the features. This confirms that the primary macro-relationships within the dataset are largely linear.

However, the test partition reveals a distinct performance asymmetry: the model maintains high anomaly precision (**85.33%**) but struggles with a more moderate recall (**71.86%**). While the linear PCA transformation effectively isolates prominent outliers, it strips away the subtle, non-linear interactions required to flag stealthy attack variants. This linear bottleneck explains why the non-linear feature extraction performed by the Autoencoder (Section 4.4) achieves a significantly higher anomaly recall (**88.21%**).

4.6 Comprehensive Method Comparison

Table 9 aggregates the final performance across all evaluated models on the test partition.

Anomaly Detection Method	Train F1-Macro	Test F1-Macro	Test Precision (A)	Test Recall (A)
Original OC-SVM ($\nu = 0.05$)	0.9026	0.7477	0.8084	0.8258
PCA + OC-SVM ($\nu = 0.01$)	0.8835	0.7377	**0.8533**	0.7186
AE Reconstruction Error	0.9132	0.7547	0.7994	0.8669
AE Bottleneck + OC-SVM ($\nu = 0.1$)	0.8300	**0.7795**	0.8168	**0.8821**

Table 9. Comprehensive comparison of anomaly detection methods across metrics.

The **AE Bottleneck + OC-SVM** achieves the highest test F1-Macro (**0.7795**), proving that deep, non-linear compression isolates complex boundaries that linear PCA (0.7377) and raw high-dimensional spaces (0.7477) fail to capture.

A detailed look reveals a classic architectural trade-off: while the linear PCA baseline maximizes anomaly precision (**85.33%**), its rigid boundaries suffer from a low recall (**71.86%**), missing stealthy, non-linear attack variations. Conversely, the optimal autoencoder pipeline achieves the best anomaly recall overall (**88.21%**) with a minimal precision penalty (**81.68%**), confirming that deep feature extraction handles complex attack signatures far more effectively.

5 UNSUPERVISED ANOMALY DETECTION AND INTERPRETATION

This section explores purely unsupervised clustering approaches (K-Means and DBSCAN) for anomaly detection, evaluating their ability to separate attack types without label supervision.

5.1 K-Means Clustering

We applied K-Means with $k = 4$ clusters on the full training set. Following the assignment assumptions, we presume that a domain expert may know the approximate number of common traffic categories, but lacks access to the specific attack labels associated with individual samples. The resulting clusters were therefore generated in a completely unsupervised manner and evaluated afterwards using the ground-truth labels.

Cluster	Size	Silhouette	Dominant Class	Purity	Composition
0	1,560	0.261	probe (54%)	Mixed	probe 54%, normal 28%, dos 18%
1	1,422	0.753	dos (100%)	Pure	dos 100%
2	9	0.392	normal (100%)	Pure	normal 100%
3	12,072	0.461	normal (85%)	Mixed	normal 85%, probe 8%, dos 6%, r2l 1%

Table 10. K-Means cluster composition and quality ($k = 4$, training set).

The average silhouette score of 0.468 indicates moderate cluster separation overall. **Cluster 1** exhibits a high silhouette score of 0.753 because DoS floods generate a distinctly dense region in feature space, driven by high SYN error rates and connection counts. **Cluster 0** has the lowest silhouette (0.261), corresponding to an ambiguous boundary region that mixes probe, normal, and some DoS traffic. Conversely, the algorithm struggled heavily with stealthy attacks: almost all 145 R2L attacks were absorbed into the massive, normal-dominated **Cluster 3**. This suggests that R2L attacks share many mathematical characteristics with normal traffic under the chosen feature representation and distance assumptions, preventing a variance-minimizing algorithm like K-Means from partitioning them effectively.

Cluster 2 contains only 9 samples, all labelled as normal traffic. Although very small, this cluster is still informative because it identifies a group of highly atypical normal observations that are separated from the rest of the dataset. Such clusters are particularly relevant for the subsequent DBSCAN analysis, where pure-normal clusters can be used to estimate density parameters.

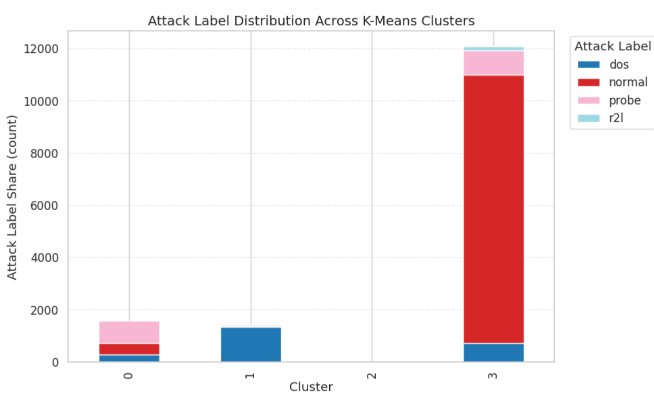

Fig. 6. K-Means ($k = 4$) cluster composition by traffic type.

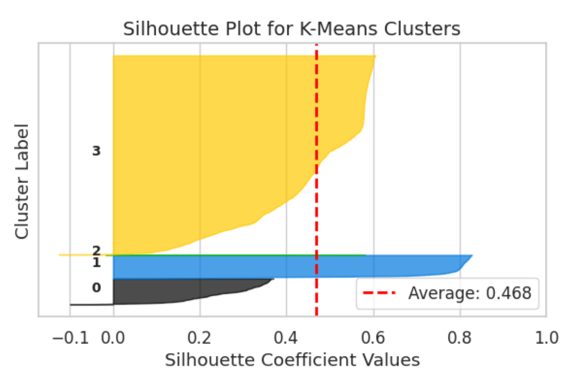

Fig. 7. K-Means ($k = 4$) silhouette coefficient distribution.

Using majority voting to assign binary labels to clusters, K-Means achieves a training F1-Macro of **0.7954**. However, this post-hoc labelling requires access to ground-truth labels for evaluation and does not generalize to unseen data, limiting its practical utility as a standalone IDS.

5.2 t-SNE Visualization

We applied t-SNE to project the training data into 2D, testing perplexity values of 5 (local focus), 30 (balanced), and 100 (global). **Perplexity = 30** provides the best balance between local and global structure, preserving meaningful class separability without over-fragmenting the data.

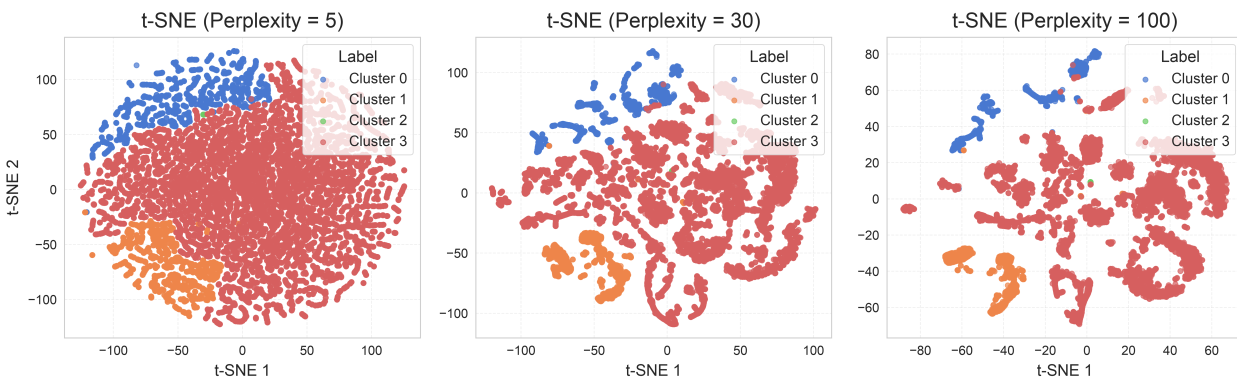

Fig. 8. t-SNE at three perplexity values. Perplexity = 5 over-fragments the data into micro-clusters; perplexity = 100 merges distinct attack islands; perplexity = 30 preserves both local separation and global structure.

The side-by-side comparison in Figure 9 reveals that K-Means quite well separates the DoS cluster (Cluster 1 aligns cleanly with the DoS island in t-SNE space), but **Cluster 3 absorbs diverse traffic types**. Most notably, R2L samples appear largely embedded within regions dominated by normal traffic in the t-SNE projection, providing a visual explanation for the difficulty encountered by K-Means when attempting to isolate them.

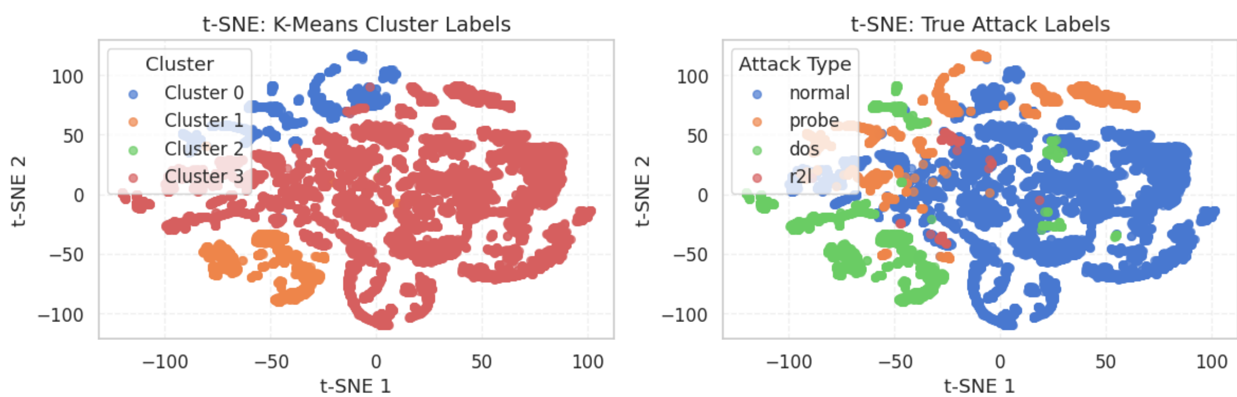

Fig. 9. t-SNE (perplexity = 30). Cluster 1 aligns cleanly with DoS (bottom-left), but Cluster 3 absorbs both probe and R2L traffic.

5.3 DBSCAN Clustering

DBSCAN identifies dense clusters and labels the remaining points as noise (-1), which we interpret as potential anomalies. We utilized Cosine distance throughout this clustering analysis to emphasize similarities in traffic profiles rather than absolute traffic volume, which can vary substantially across connections. This directional alignment provides a alternative neighborhood definition for DBSCAN than raw magnitude differences.

Following the assignment guidelines, we first inspected the K-Means solutions across various parameter configurations to identify the smallest cluster composed exclusively of benign traffic samples. This optimization heuristic yielded a baseline target value of `min_pts` = 7 (derived from Cluster 2 in Table 10). However, such a small value is likely to generate clusters around very small groups of observations, including isolated normal outliers. To investigate more robust and conservative density definitions, we extended our exploratory parameter sweep to include `min_pts` $\in \{7, 100, 500, 1600\}$. For each setting, the corresponding operational ϵ was determined using the sorted

k-distance graph elbow rule, locating the inflection point at the 70th percentile of sorted distances (yielding $\epsilon = 0.635$ for min_pts = 1600, and $\epsilon = 0.260$ for min_pts = 500).
The behavior of DBSCAN changes considerably as the density threshold increases. Lower values of min_pts generate a highly fragmented space with numerous micro-clusters, classifying a significant volume of sparse, legitimate observations as noise. Conversely, elevated parameters progressively force the data coordinates to coalesce into massive, overarching spatial domains.

Among the explored configurations within our parameter sweep, min_pts = 500 ($\epsilon = 0.260$) provides the most balanced and interpretable trade-off between cluster granularity and noise isolation. This specific configuration populates exactly four distinct data clusters and captures the highest overall quantitative metric with a training F1-Macro score of 0.4874. Although the resulting noise fraction remains heavily dominated by benign traffic anomalies (70.7%), it provides the most interpretable compromise between cluster granularity and noise detection among the tested configurations.

min_pts	ϵ	Clusters	Noise %	F1-Macro	Noise: Normal	Noise: Anomaly
7	0.064	40	3.2%	0.4285	80.9%	19.1%
100	0.053	17	21.8%	0.4707	76.0%	24.0%
500	0.260	4	14.2%	0.4874	70.7%	29.3%
1600	0.635	1	2.0%	0.4466	52.5%	47.5%

Table 11. DBSCAN results for different values of min_pts.

Adjusting the density parameter up to min_pts = 1600 changes the noise cluster's composition, moving it closer to an even split with malicious instances (47.5% true anomalies). However, this marginal shift comes at a critical structural cost, collapsing almost the entire feature space into a single massive cluster (Cluster 0), as detailed in Table 12.

Attack Label	Noise (-1)	Cluster 0	Total
Normal	159	10,599	10,758
DoS	138	2,192	2,330
Probe	1	1,829	1,830
R2L	5	140	145
Total	303	14,760	15,063

Table 12. DBSCAN matrix breakdown under a single-cluster collapse (min_pts = 1600, $\epsilon = 0.635$).

For this dataset, the density structure mapped by DBSCAN does not align cleanly with the actual distinction between normal and malicious traffic. This limitation manifests in three main ways:

- **DoS Floods:** Because volumetric flood attacks consist of thousands of identical, rapid packets, DBSCAN groups them into a dense cluster rather than identifying them as noise anomalies.
- **Sparse Normal Outliers:** Atypical but benign connections (e.g., with unusual session durations or rare services) appear isolated and are falsely flagged as malicious noise.
- **R2L Intrusions:** Low-volume privilege-escalation exploits closely mimic normal behavior, remaining hidden inside normal clusters.

Overall, these results show that density-based clustering alone cannot reliably separate normal and anomalous traffic here. While DBSCAN helps analyze the structure of the network data, it is not suitable as a standalone intrusion detection system.